

Online Shoppers Purchasing Intention

Predicting session-level conversion from in-session browsing behaviour

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IE 7275: Data Mining in Engineering

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Outline

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- 3 Data and preparation
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- 5 Modelling
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- An end-to-end data mining study on a real-world dataset, from raw records to decision-relevant findings.
- Exploratory analysis to understand the data, surface structure, and guide modelling choices.
- A set of classification models compared under realistic conditions, including class imbalance.
- Meaningful interpretation of the results, rather than mere performance evaluation.

Problem setting

- Many small retailers operate online to reduce overhead costs.
- Online data are abundant but harder to read than a physical storefront, where an owner can directly infer intent from visual cues.
- Without a sound understanding of the data, small retailers may misallocate limited budgets toward ineffective campaigns and site designs.

Question

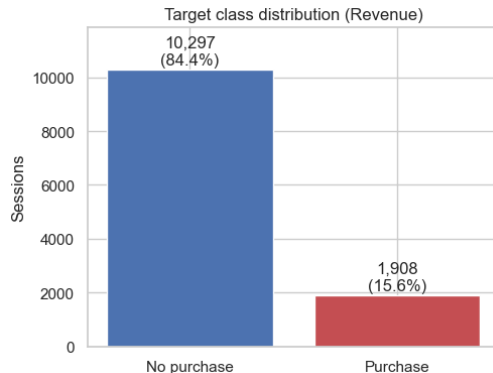
Can session behaviour alone tell us who is about to buy, and where to make changes to the storefront?

- **Primary objective:** To predict the binary target `Revenue` from session activity.
- **Secondary objective:** To rank predictors by contribution to purchase intent to identify elements the retailer can change to improve conversion rate.

Because 84.4% of sessions do not convert, a naive model always predicting “no purchase” scores 84.4% accuracy while finding no buyers, and thus accuracy is not a priority.

Data source and description

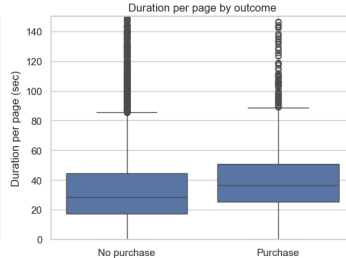
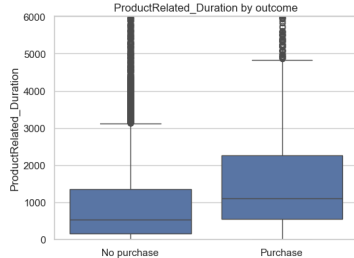
- **Source:** Online Shoppers Purchasing Intention, UCI ML Repository (Sakar, 2018). Bounce/exit rates and page value from Google Analytics.
- **Records:** 12,330 sessions over one year, of which 12,205 are unique.
- **Variables:** 17 predictors, 1 binary target. Session counts and durations are stored as numerics, alongside categoricals (month, visitor type, region, traffic, OS, browser).
- Notably, there is a **heavy class imbalance**: Only 15.6% of sessions convert into sales.



Data preparation

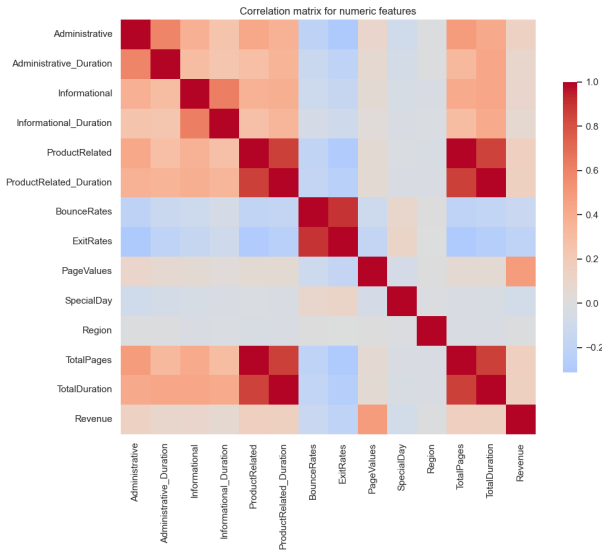
- The data contained no missing values.
- 125 **duplicate rows were removed** to prevent train/test leakage, leaving 12,205 sessions.
- Long sessions were flagged by Tukey's rule ($3 \times \text{IQR}$ on duration-per-page): 285 outlier sessions (2.34%) converted at 10.9%, below the 15.6% average, suggesting idle time rather than active shopping.
- Nonetheless, **all unique rows were retained**. Tree models are robust to outliers, and our logistic models are protected by standardization and shrinkage. The low conversion among long sessions is in and of itself a signal.

Behaviour by outcome



- Converting sessions spend more time viewing product pages.
- They also spend **more time per page**, not just more pages overall.
- Engagement depth rather than volume separates buyers from non-buyers.

Correlation and multicollinearity



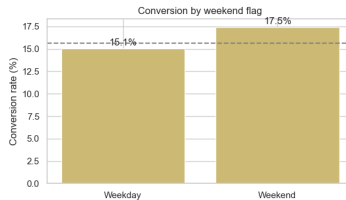
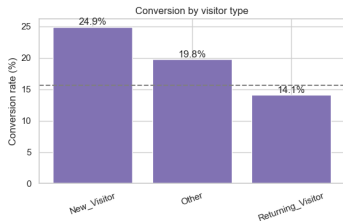
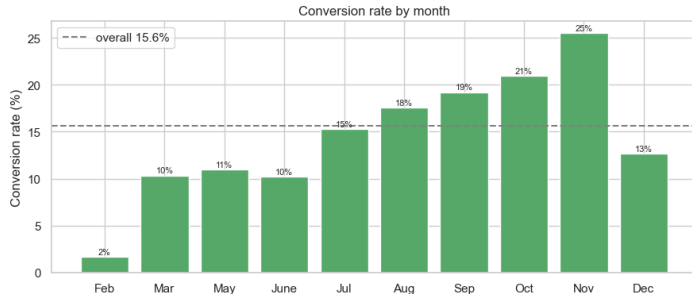
Strongest pairs:

- ProductRelated and TotalPages: 0.997
- ProductRelated_Duration and TotalDuration: 0.994
- BounceRates and ExitRates: 0.902

PageValues is the predictor most correlated with the target.

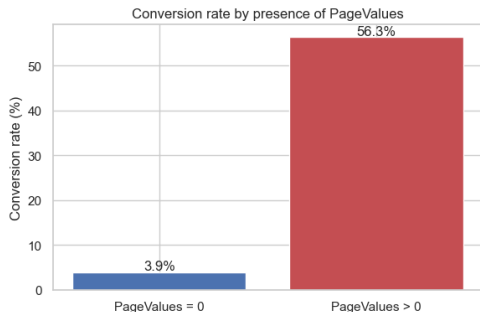
Regularization and trees absorb the collinearity, so **PCA is not used** and ease of interpretability is maintained.

Conversion by segment



- No observations from Jan or Apr were present. Nov conversion is more than double that of those from Feb through Jun.
- New visitors converted at roughly twice the rate of returning ones.
- Weekend shoppers were minimally more likely to convert over weekday shoppers.

Page value as a dividing line



- 22.4% of sessions reach a value-bearing page.
- Those sessions convert at 56.3%.
- Sessions with PageValues = 0 convert at 3.9%.

PageValues, a Google Analytics engineered metric, carries most of the separating power for this dataset. However, its nature matters for how it can be used (revisited in results).

Data mining task and approach

- **Task:** Supervised binary classification, optimised for the minority (purchase) class.
- **One-hot encoding** of categoricals expanded 19 to 75 columns. Logistic models used standardized features for the linear models, while raw features were kept for trees.
- The data were split 80/20 **stratified** on the target, yielding 15.6% positive in both partitions.
- **Randomized oversampling** was done in the training set for the baseline logit. For the other models, `class_weight / scale_pos_weight` was passed in CV to avoid leakage between folds.
- Models were tuned with **10-fold stratified CV, scored on F1**.

Models tested

- **Logistic regression (baseline):** Unregularized as an interpretable reference point.
- **LASSO, `lambda.min`:** L_1 penalty tuned over 50 values of C ; embedded feature selection.
- **LASSO, `lambda.1se`:** Most parsimonious model within one standard error of the best; dropped to 12 features retained versus 34 for `lambda.min`.
- **Random forest:** Tuned on tree count and depth.
- **XGBoost:** Gradient-boosted trees, `scale_pos_weight` set to the negative/positive ratio.
- **Soft-vote ensemble:** Averaged probabilities of LASSO, random forest, and XGBoost.

Performance evaluation

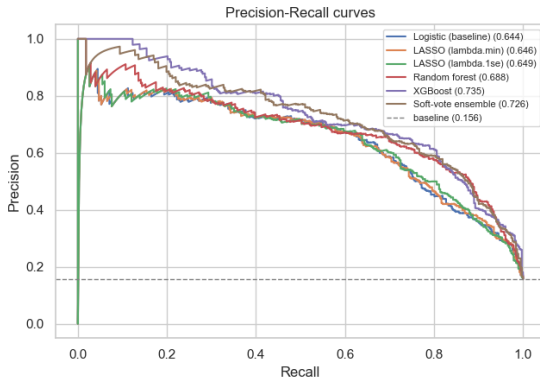
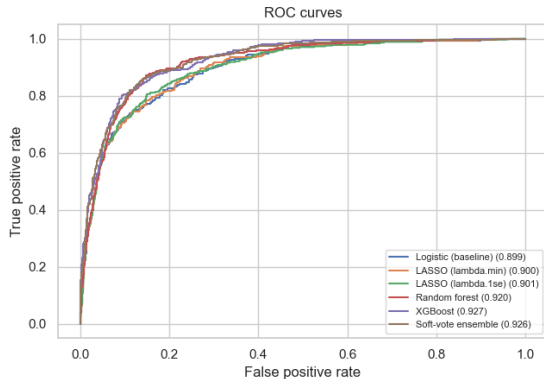
Ranked by F1 on the held-out test set ($n = 2,441$).

Model	F1	Precision	Recall	PR-AUC	ROC-AUC
XGBoost	0.693	0.618	0.791	0.735	0.927
Soft-vote ensemble	0.673	0.591	0.780	0.726	0.926
Random forest	0.669	0.588	0.775	0.688	0.920
LASSO (<code>lambda.1se</code>)	0.627	0.537	0.751	0.649	0.901
LASSO (<code>lambda.min</code>)	0.608	0.503	0.767	0.646	0.900
Logistic (baseline)	0.602	0.495	0.770	0.644	0.899

- F1 and PR-AUC are particularly important for our imbalanced dataset.
- XGBoost topped the charts on every metric. For additional interpretability based on coefficients, the `lambda.1se` LASSO model maintained most of the performance and generalized well.

Ranking quality across thresholds

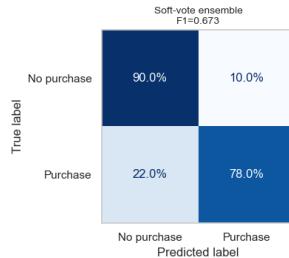
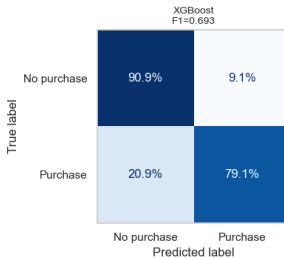
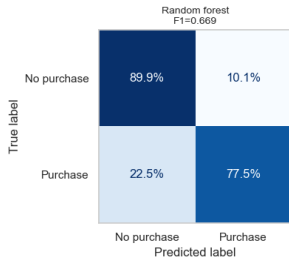
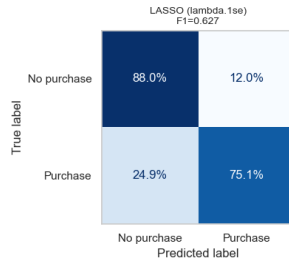
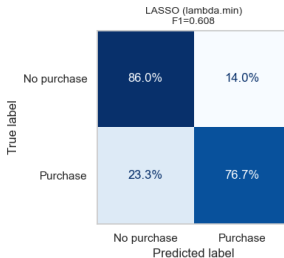
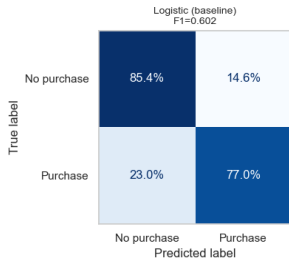
Areas under curves



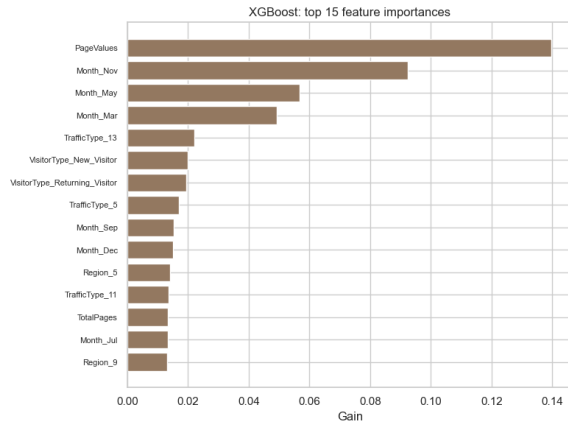
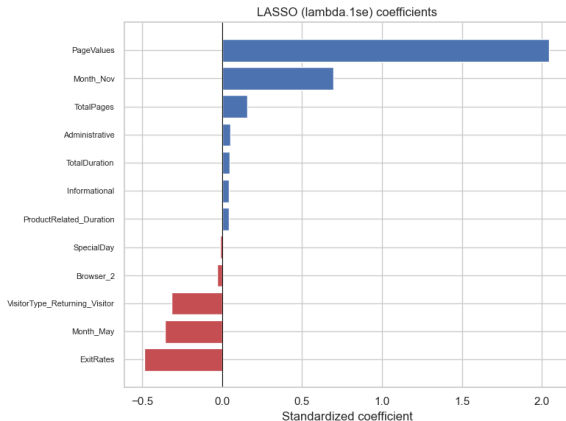
- Separation in PR-AUC (relevant for imbalance) clearly favoured XGBoost (max 0.091 over baseline)
- Differences in AUC-ROC were relatively minor in comparison (max 0.028 over baseline)

Confusion matrices at threshold 0.5

Row-normalized confusion matrices



Feature importance across models



PageValues dominated across all models, with exit rate, month (November), and visitor type following.

- PageValues is the strongest predictor by a wide margin, but it is built from past transactions. It detects high-intent sessions in real time, and can not be adjusted to improve conversion.
- Exit and bounce rates are more actionable, as they show which pages visitors are leaving from.
- Time of year matters, as conversion rates increase in the fall, peaking in November at more than double the spring rate.
- New visitors convert at nearly twice the rate of returning ones, likely arriving through high-intent channels such as search results or targeted advertisements.

- Use `PageValues` as a signal to flag high-intent sessions live and improve conversion (i.e. through targeted coupons), rather than using it as a target.
- Use exit and bounce rates to locate drop-off points, the most directly actionable means to improve retention.
- Concentrate promotional spend on the seasonal run-up to December, as customers steadily convert more from August through November.
- Prioritize new visitors upon arrival, rather than attempting to increase traffic further.

- A small e-tailer gains a direction for **allocating their limited budget** for advertising and design.
- Recall near 0.79 at the default threshold means **most buyers are flagged**; the threshold can be further tuned to the cost of a missed sale versus a wasted advertisement.
- **Interpretability is preserved** by the parsimonious 12-feature LASSO while the tree importances help provide an explainable story to a non-technical owner or operator.
- The **findings may be transferrable** to comparable online storefronts, subject to local validation.

- In-session behaviour predicts conversion reasonably well:
 - XGBoost reaches F1 0.693, PR-AUC 0.735.
- A compact 12-feature model recovers most of the performance, so interpretability costs little.
- PageValues detects intent; exit and bounce rates are where the store can act.
- Seasonality and visitor type sharpen targeting.

Future extensions

- **A/B testing** of storefront changes guided by high-exit pages, to test causality and check for improvement.
- **Threshold and cost analysis** tying the decision boundary to revenue per conversion and intervention cost.
- **Calibration and monitoring** of predicted probabilities for live deployment (i.e. real-time coupon targeting).
- **Collection of richer data** such as clickstream sequence, session recency, and channel-level detail for further analysis.

Sakar, C. (2018). *Online Shoppers Purchasing Intention Dataset*. UC Irvine. doi:10.24432/c5f88q

Thank you!

Questions?